

1

$$\frac{dy}{dx} = -\frac{4x^3}{4y^3} = -\frac{x^3}{y^3} \quad (\text{B.1}) \quad -2\sqrt{2}$$

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = -\frac{3x^2y^3 - x^3 \cdot 3y^2 y'}{(y^3)^2} \quad (\text{B.1}) = 2\sqrt{2} \cdot 3 \cdot (-2\sqrt{2}) - 6 = -30 //$$

2

$$\lim_{x \rightarrow \infty} \eta \left(1 - \frac{x^2}{2n} + \frac{x^3}{n^2} \right) \cong -\frac{x^2}{2n} + \frac{x^3}{n^2} \quad (\because \ln(1+x) \cong x)$$

$$\Rightarrow e^{\lim_{x \rightarrow \infty} \eta \left(1 - \frac{x^2}{2n} + \frac{x^3}{n^2} \right)} = e^{-\frac{x^2}{2}} = f(x).$$

$$f(x) = x e^{-\frac{x^2}{2}} \Big|_{x=2} = -2e^{-2} //$$

3

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2t^2-2}{2t} \Big|_{t=1} = \frac{1}{2}$$

$$\frac{d^2y}{dx^2} = \frac{1}{dx} \left(\frac{dy}{dx} \right) = \frac{d\left(\frac{dy}{dx}\right)}{dx} = \frac{1}{\frac{dx}{dt}} \cdot \frac{d\left(\frac{dy}{dx}\right)}{dt} \Big|_{t=1}$$

$$= \frac{1}{2} \cdot \frac{4-2}{4} = \frac{1}{4} //$$

4

$$\int_{-1}^3 x dx - a \int_{-1}^3 |x| dx = 4 - 5a = 10$$

$$\frac{1}{2} \cdot (9-1) - \frac{1}{2} (1+3 \times 3) \quad \therefore a = -\frac{6}{5} //$$

5

$$\int_0^{2\sqrt{\pi}} f(s) ds = \int_0^{2\sqrt{\pi}} \left(\frac{d}{ds} \int_0^{s^2} 4s^2 dt \right) ds.$$

$$4s \cdot 4s^2(s^2) - 2s \cdot 4s^2(s^2).$$

$$I_1: \int_0^{2\sqrt{\pi}} 4s \cdot 4s^2(s^2) ds$$

$$2s^2 \leq u \Rightarrow 4s ds = du \Rightarrow \int_0^{8\pi} 4s^2 u du = 4 \cdot 4 \cdot \frac{1}{2} \cdot \frac{\pi}{2} = 4\pi.$$

$$I_2: \int_0^{2\sqrt{\pi}} 2s \cdot 4s^2(s^2) ds.$$

$$s^2 \leq u \Rightarrow 2s ds = du \Rightarrow \int_0^{4\pi} 4s^2 u du = 2 \cdot 4 \cdot \frac{1}{2} \cdot \frac{\pi}{2} = 2\pi.$$

$$\therefore I_1 - I_2 = 2\pi //$$

6

* 중요

$$\therefore \frac{1}{12} \cdot 1^2 + \frac{1}{2} = \frac{7}{12} //$$

7

$$\lim_{x \rightarrow \infty} \frac{x^n}{n!} = e^x - (1+x) \Big|_{x=1} = 6 - \ln 7 //$$

8

$$f(x) = (\sin x)^3 = \left(x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \dots \right)^3$$

$$\Rightarrow \left(x - \frac{x^3}{6} + \frac{x^5}{120} \right) \left(x - \frac{x^3}{6} + \frac{x^5}{120} \right) \left(x - \frac{x^3}{6} + \frac{x^5}{120} \right).$$

3	3	5	(x 3)
3	3	1	(x 3)

$$\therefore 3 \cdot \left(\frac{1}{120} + \frac{1}{36} \right) = \frac{1}{40} + \frac{1}{12} = \frac{3+10}{120} = \frac{13}{120}$$

$$\therefore f^{(11)}(0) = \frac{13}{120} \cdot 11! = 13 \cdot 9 \cdot 6 = 546 //$$

9

$$\frac{dm}{dn} = 2 \Rightarrow \ln m = 2 \ln n \Rightarrow \frac{1}{m} \ln m = \ln n.$$

$$\therefore \lim_{n \rightarrow \infty} \left(\frac{\ln m}{n} \right)^{\frac{1}{n} \cdot \ln m} = \lim_{n \rightarrow \infty} \left(\frac{1}{n} \right)^{\frac{1}{n} \cdot \frac{1}{n}} = \frac{1}{e} //$$

10

$$\begin{cases} x = u+v \\ y = 3u^2 \dots (2, 3, 0) \Rightarrow u=1, v=1. \\ z = u-v \end{cases}$$

$$r(u, v) = (u+v, 3u^2, u-v)$$

$$r_u = (1, 6u, 1) \Big|_{u=1} = (1, 6, 1), \quad r_v = (1, 0, -1)$$

$$r_u \times r_v = \begin{pmatrix} 6 & 1 & 6 \\ 0 & -1 & 0 \end{pmatrix} = \langle -6, 2, -6 \rangle$$

$$\therefore d: 3x - y + 3z = 3$$

$$\therefore d = \frac{| -3 |}{\sqrt{9+1+9}} = \frac{3}{\sqrt{19}}$$

11

$$V = \frac{1}{2} \begin{vmatrix} 2 \sec t & -1 \\ \sec t & -1 \\ 2 \sec t & 0 \end{vmatrix} = 2(2 \sec t - 1) - \sec t(4 + \sec t)$$

$$= \frac{1}{2}(-4 \sec t - 2 - 4 \sec t - \sec^2 t) = \frac{1}{2}(-2 - \sec^2 t)$$

$$\therefore V = \frac{1}{2}|-2 - \sec^2 t| \rightsquigarrow V_{\min} = \frac{1}{2}$$

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$$W = \begin{pmatrix} x \\ y \\ z \end{pmatrix} t$$

$$W_t = W_x \cdot \frac{dx}{dt} + W_y \cdot \frac{dy}{dt} + W_z \cdot \frac{dz}{dt}$$

$$t = \frac{2}{3}\pi \Rightarrow x = \rho^{\frac{4}{3}} \cdot \frac{1}{2}, \quad y = -\frac{1}{2}, \quad z = \frac{\sqrt{3}}{2}$$

$$W_t \left(t = \frac{2}{3}\pi \right) = 2\rho^{\frac{4}{3}} \cdot 2\rho^{\frac{4}{3}} - 1 \cdot \left(-\frac{\sqrt{3}}{2}\right) + 2\sqrt{3} \cdot \left(-\frac{1}{2}\right)$$

$$= 4\rho^{\frac{8}{3}} - \frac{\sqrt{3}}{2}$$

13

$$g(t) = \phi(r(t)) \cdot r'(t) \Big|_{t=0} = \phi(r(0)) \cdot r'(0) = u \cdot v = 0.$$

$$\therefore u \perp v$$

$$|u \cdot v| = \sqrt{3+1} = 2$$

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$$x = \frac{r}{\sqrt{5}} \cos \theta, \quad y = r \sin \theta \quad (0 < r \leq 1, 0 \leq \theta \leq 2\pi)$$

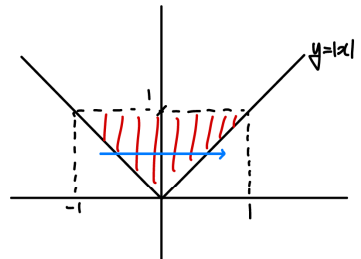
$$\Rightarrow 3x^2 - y^3 = \frac{r^3}{\sqrt{5}} \cos^2 \theta - r^3 \sin^3 \theta = \frac{r^3}{\sqrt{5}} (\cos^2 \theta - \sin^3 \theta)$$

$$g(\theta) = \frac{3}{\sqrt{5}} \cos^2 \theta - \sin^3 \theta = 3 \sin \theta \cos \theta \left(-\frac{1}{\sqrt{5}} \cos \theta - 1 \right) = 0$$

$$\therefore \theta = 0, \quad \theta = \frac{\pi}{2}, \quad \theta = \pi, \quad \theta = \frac{3\pi}{2}, \quad \theta = 2\pi \quad \& \quad \cos \theta = -\frac{1}{\sqrt{5}} \Rightarrow \sin \theta = \frac{2}{\sqrt{5}}$$

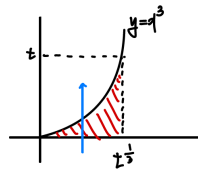
$$\therefore M=1, \quad m=-1 \Rightarrow M-m=2$$

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$$\Rightarrow \int_0^1 \int_{-y}^y e^{y^2} dx dy = \int_0^1 2ye^{y^2} dy = \frac{e-1}{2}$$

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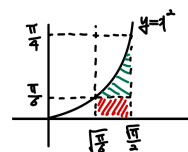


$$f(t) = \int_0^t \int_0^{t^3} (20 + 22 \sin(\frac{\pi}{2} x^{2004})) dx dy$$

$$(20t^3 + 22t^3 \sin(\frac{\pi}{2} t^{2004}))$$

$$f(t) = \frac{1}{3} t^{\frac{2}{3}} \cdot (20t + 22t \cdot \sin \frac{\pi}{2} t^{\frac{2004}{3}}) \xrightarrow{t=1} \frac{1}{3}$$

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$$\Rightarrow \int_{\frac{\pi}{8}}^{\frac{\pi}{4}} \int_0^{\frac{1}{\sqrt{2}}} \frac{y \cos(x)}{y^3} dy dx = \int_{\frac{\pi}{8}}^{\frac{\pi}{4}} \frac{1}{2} \cdot \frac{\cos(x)}{y^2} dy dx = \frac{1}{4} \cdot [\sin(x)]_{\frac{\pi}{8}}^{\frac{\pi}{4}}$$

$$= \frac{1}{4} \left(\frac{\sqrt{2}}{2} - \frac{1}{2} \right) = \frac{1}{8} (\sqrt{2} - 1)$$

18

$$S = \iint \sqrt{1+4x^2+4y^2} dA = 2 \cdot \int_0^{\frac{\pi}{2}} \int_1^{\sqrt{3}} \sqrt{1+4r^2} dr d\theta$$

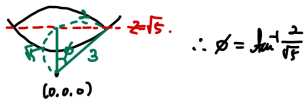
$$= \pi \cdot \left[\frac{1}{\sqrt{2}} (1+4r^2)^{\frac{3}{2}} \right]_1^{\sqrt{3}} = \frac{\pi}{\sqrt{2}} (13\sqrt{13} - 5\sqrt{5}) //$$

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$$\frac{4\pi}{22+44-22=44} \iiint \frac{1}{\sqrt{1+x^2+y^2+z^2}} dV = 4 \int_0^1 \int_0^{\frac{\pi}{4}} \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt{1+r^2}} r^2 dr d\theta dy = \frac{1}{\sqrt{2}} //$$

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$$x^2y^2z^2=9 \text{ \& } x^2=z^2 \Rightarrow x^2y^2=z^2 : z=\sqrt{5}$$



$$S = \iint \underbrace{\sqrt{1+z^2}}_{\frac{4}{3}\sin\theta} d\theta d\phi = 9 \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{15}} \sin\theta d\theta d\phi = 18\pi \cdot \left[-\cos\theta \right]_0^{\frac{\pi}{2}}$$

$$= 18\pi \cdot \left(1 - \underbrace{\cos\left(\frac{\pi}{2}\right)}_{\frac{\sqrt{15}}{3}} \right) : \sin\alpha = \frac{2}{\sqrt{5}} \Rightarrow \cos\alpha = \frac{\sqrt{5}}{3}$$

$$= 18\pi \left(1 - \frac{\sqrt{5}}{3} \right) = \frac{6\pi}{1} (3 - \sqrt{5}) //$$